INCH-POUND
MIL-DTL-19264D(OS)
26 July 2016
SUPERSEDING
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23 October 2009

DETAIL SPECIFICATION

PROPELLANT, BS-NACO, FOR PROPELLING CHARGES FOR NAVY 5-INCH GUNS

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the minimum requirements for Butyl Stearate Navy Cool (BS-NACO) propellant for use in propelling charges used in Navy 5-inch guns (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-G-155	-	Graphite, Dry (for Use in Ammunition)
MIL-DTL-244	-	Nitrocellulose
MIL-E-255	-	Ethyl Centralite (Carbamite)
MIL-DTL-18618	-	Lead Carbonate, Basic, Dry (for Ordnance Use)
MIL-B-21465	-	Butyl Stearate, Normal
DTL-WS-35709	-	Potassium Sulfate (for Ordnance Use)

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-652	-	Propellants, Solid, for Cannons Requirements and Packing
DOD-STD-2101	-	Classification of Characteristics

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

(Copies of DTL-WS-35709 are available from the Department of the Navy, Indian Head Division, NSWC IHEODTD, Code E14, Surface Systems Engineering Branch, 3703 Benson Road, Indian Head, MD 20640-5085, OFFICIAL BUSINESS, or by email request to ihdiv_standardization_team@navy.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (CAGE CODE 10001) DRAWINGS

2814087	-	General Arrangement (MK 19 Mod 2)
3208375	-	Charge, Propelling, 5 Inch, 54 Caliber: Full Mark 67 Mod 3

(Copies of these NAVSEA drawings are available from U.S. Navy Major Caliber Propelling Charge AEA, Naval Surface Warfare Center Indian Head Explosives Ordnance Disposal Technology Division Picatinny Detachment, ATTN: G2, Picatinny, NJ 07806-5000.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3.1) in accordance with 4.2.

3.2 Chemical composition (M101). The propellant and the propellant ingredients shall be in accordance with their respective specifications listed in table I (see 4.5.1 for test methods). M101 indicates this is a major requirement that shall be interpreted in accordance with 4.4. The classification of 4.4 shall apply for similarly designated critical (Cx) and major (M1xx) requirements throughout this specification.

3.3 Grain form and dimensions (M102).

3.3.1 Perforations. The grains shall be cylindrical with seven, evenly spaced, longitudinal perforations; one in the center of the grain and six at the vertices of a symmetrical hexagon (see 4.5.2, 4.5.3, and 6.3.1).

3.3.2 Length. The length (L) to outer diameter (D) ratio for the grains shall be between 2.10 and 2.50 (see 4.5.2, 4.5.3, and 6.3.1).

3.4 Glaze. The propellant shall be glazed with graphite, MIL-G-155, Grade IV (see 4.5.3).

3.5 Integrity. The propellant grains shall be without cracks or incipient cracks between the perforations or between the perforations and the outside edge of the grain (see 4.5.3).

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TABLE I. BS-NACO composition.

Material	Specification	Percent by Weight
Nitrocellulose, Grade E, Type I	MIL-DTL-244 1/	93.55 (nominal)
n-Butyl Stearate	MIL-B-21465 2/	3.00±0.30
Ethyl Centralite, Class 2	MIL-E-255	1.20±0.20
Basic Lead Carbonate	MIL-DTL-18618 3/	1.00±0.20
Potassium Sulfate, Type I	DTL-WS-35709	1.25±0.30
Moisture	---	1.00 (min) to 3.00 (max)
Total Volatiles	MIL-STD-652	5.0 (max)
Graphite, Grade IV	MIL-G-155	0.10±0.05

NOTES:

1/ The nitrocellulose shall have an average nitrogen content of 12.00±0.10 percent and shall have a viscosity of 5±2 seconds. The required nitrogen content may be obtained by blending nitrocellulose containing 11.90 to 12.20 percent nitrogen. Average nitrogen content determination shall be made only when specifically requested (see 6.2).

2/ The n-Butyl Stearate specific gravity shall be determined at 25 °C and the sample weight shall be 2.5 to 3.2 grams for iodine test.

3/ The compatibility tests for lead carbonate with Cordite-N propellant shall not be required.

3.6 Stability (C1). The methyl violet test paper shall not fade to a salmon-pink color in less than 40 minutes and the propellant shall not self-initiate in less than 5 hours (see 4.5.4).

3.7 Production packing depth (M103). The production packing depth (PPD) of the assessed charge (AC) weight shall be at least 3.5 inches (see 4.5.5).

3.8 Gun ballistic assessment requirements.

3.8.1 Velocity (M104). The velocity of the test propellant (TP) shall match the nominal average muzzle velocity of the reference propellant (RP) (see 6.3.2).

3.8.2 Pressure (M105). The maximum chamber pressure for the assessed charge weight of the TP shall not exceed 55,100 pounds per square inch (psi) (see 4.5.6.2).

3.8.3 Velocity variation (M106). The standard deviation for projectile muzzle velocity shall not exceed 10 feet per second (ft/s) when the 5-inch, 54-caliber (5"/54) barrel is used for ballistic testing. When the 5-inch, 62-caliber (5"/62) barrel is used, the standard deviation for projectile muzzle velocity shall not exceed 9.5 ft/s (see 4.5.6.2.2.5).

3.8.4 Pressure variation (M107). The standard deviation for the maximum chamber pressure (PSD) shall not exceed 1,700 psi (see 4.5.6.2.3.3).

3.8.5 Propellant interface.

3.8.5.1 Temperature coefficient. The temperature coefficient of the TP shall not deviate from that of the RP by more than ±0.2 (ft/s)/°F (see 4.5.6.3).

3.8.5.2 Eject time. The eject time of the TP shall not exceed 50 milliseconds (see 4.5.6.3.2).

3.8.5.3 Pressure waves. The individual maximum initial negative pressure differential ($-\Delta P_i$) shall not exceed 5,000 psi (see 4.5.6.3.2.1).

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3.9 Closed bomb.

3.9.1 Relative force. The average relative force (RF) value shall be within ±2.0 percent of that of the RP when tested in accordance with 4.5.7 and calculated from 4.5.7.5.1.

3.9.2 Relative quickness. The overall propellant shall provide a relative quickness (RQ) within ±3 percent of the RP when tested in accordance with 4.5.7 and calculated from 4.5.7.5.2. Each sub-lot shall provide an RQ within ±15 percent of the RP when blending of smaller sub-lots will occur to make the overall propellant.

3.9.3 Relative vivacity. The relative vivacity (RV) values shall lie between 0.95 and 1.05 across the normalized pressure range of 0.20 (P/P_{max}) and 0.60 (P/P_{max}) when the RP and TP are tested in accordance with 4.5.7 and the RV is assessed in accordance with 4.5.7.5.3.

3.10 Workmanship. The propellant shall be a product free from foreign material, uniform in quality, and manufactured in a manner to assure compliance with requirements of this specification when examined in accordance with 4.5.3.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When required (see 3.1), first article inspection shall be performed on propellant that has been produced with equipment and procedures normally used in production. First article inspection shall consist of the tests specified in table II at a location acceptable to the Government, as specified (see 6.2). More recent or updated methods of testing may be used if approved by NAVSEA, as specified (see 6.2). Failure of the sample to meet the requirements of this specification shall be cause for rejection of the first article sample.

TABLE II. Description of inspections.

Characteristic	Requirement	Test Method
Chemical Composition	3.2	4.5.1
Grain Form and Dimensions	3.3	4.5.2, 4.5.3
Glaze	3.4	4.5.3
Integrity	3.5	4.5.3
Stability	3.6	4.5.4
Production Packing Depth	3.7	4.5.5
Gun Ballistic Assessment	3.8	4.5.6
Closed Bomb	3.9	4.5.7
Workmanship	3.10	4.5.3

4.2.1 Submission. First article samples submitted for evaluation shall be in accordance with 4.3.2. When the propellant granulation:

a. Is to be established or validated, the first article sample shall consist of three pilot lots, each with a different granulation; the set of three granulations shall provide ballistic performance that brackets the requirements of 3.8. Each pilot lot shall be at least 2,000 pounds.

b. Has previously been established, the quantity of propellant for first article testing shall be at least 2,000 pounds.

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4.3 Conformance inspection. Conformance inspection shall consist of the examinations and tests of 4.5 and table II at a location acceptable to the Government, as specified (see 6.2). Failure of any sample to conform to any requirement specified herein shall be cause for rejection of the lot represented. If alternate methods and test procedures are used, prior written approval from the Government shall be obtained, as specified (see 6.2).

4.3.1 Lot size. Unless otherwise specified (see 6.2), the propellant shall be produced in homogeneous lots between 100,000 and 500,000 pounds. This can be achieved by blending smaller sub-lots; however, each sub-lot must meet the RQ requirement specified in 3.9.2.

4.3.2 Sampling. The samples shall be selected from the full propellant packing containers (see 6.6.1) so as to be representative of the lot. All samples shall be selected following the final packing of the lot. All samples shall be packed in moisture-proof, airtight containers and marked with the following in accordance with MIL-STD-129:

a. Propellant description
b. Propellant lot number
c. Number of pounds in lot
d. Manufacturer
e. Contract number

4.3.2.1 For gun performance tests. Sample size shall be at least 2,000 pounds randomly selected from full propellant packing containers (see 6.6.1) and shipped to the designated ballistic testing facility, as specified (see 6.2).

4.3.2.2 For physical characteristic tests. A 10-pound sample shall be randomly selected from full propellant packing containers (see 6.6.1) and shipped to the testing facility approved by the Government (see 6.2).

4.3.2.3 For surveillance tests. A 5-pound sample shall be randomly selected from the full propellant packing containers (see 6.6.1) and shipped to the Navy's designated surveillance activity for propellants, as specified (see 6.2).

4.4 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Requirements for critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the characteristic. Requirements that are not annotated with a classification symbol are considered to be minor characteristics.

4.5 Methods of inspection.

CAUTION

This specification covers sampling and testing of chemical,
toxic, or explosive materials, which are hazardous to personnel.

4.5.1 Chemical composition. The chemical composition of the propellant, as specified in table I, shall be determined using the methods specified in table III (see 3.2). As specified (see 6.2), the manufacturer shall certify that the ingredients for each propellant lot satisfy the requirements listed in table I. Failure to meet the requirements of 3.2 shall be cause for rejection of the lot. More recent or updated methods of testing may be used upon approval of the Government, as specified (see 6.2).

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TABLE III. Chemical composition test methods.

Material	Test Method of MIL-STD-286
Nitrocellulose	209.2.1
n-Butyl Stearate	222.1
Ethyl Centralite	202.2.3
Basic Lead Carbonate	311.1.3 or 316.1
Potassium Sulfate	310.3.1, 310.4.2, or 316.1
Moisture	103.5
Total Volatiles	103.5
Graphite	308.1.4

4.5.2 Grain dimensions. Grain dimensions shall be determined in accordance with MIL-STD-286, Method 504.1.1 or 504.6.1. The requirements of 3.3 shall be met, or the lot shall be rejected.

4.5.3 Visual examination. The propellant grains selected for the physical characteristic tests shall be visually examined prior to conducting the physical characteristic tests (see 4.5.1 and 4.5.2) to verify that:

a. Grain form conforms to the requirements of 3.3 and 3.5,

b. Grains are uniformly coated with graphite conforming to the requirements of 3.4, and

c. Workmanship conforms to the requirements of 3.10.

4.5.4 Stability. Stability shall be determined in accordance with MIL-STD-286, Method 404.1.2 (134.5 °C heat test). The requirements of 3.6 shall be met, or the lot shall be rejected.

4.5.5 PPD test. PPD tests shall be conducted during the ballistic tests (see 4.5.6). The PPD of the TP shall be assessed according to the procedure described in 4.5.5.3:

a. A preliminary PPD curve shall be developed prior to ballistic testing (see 4.5.6), based upon predicted charge weight estimates from interior ballistics (IB) modeling (see 4.5.8), and

b. A final PPD for the TP shall be determined based upon the assessed charge weight for the TP (see 4.5.6.2.2.6).

During first article inspection (see 4.2 and 6.3.1), the charge weight probe test (see 4.5.6.2.1) shall be conducted at a minimum before lot rejection becomes an option based on PPD. If lot rejection is being considered in response to the preliminary PPD results, discussion with the testing facility, Design Agent (DA), and NAVSEA shall be required, as specified (see 6.2), before making the decision to discontinue all remaining ballistic tests or determining the lot to be unacceptable.

4.5.5.1 Preliminary PPD curve. The preliminary PPD curve shall be determined using TP that has been conditioned for a minimum of 4 days at 70±3 °F. The preliminary PPD curve shall be determined by performing a series of at least three pours as described in 4.5.5.3. The curve shall be determined by performing a linear least squares fit on the resulting data.

4.5.5.2 Final PPD for the TP. After an assessed charge weight has been determined for the TP (see 4.5.6.2.2.6), the final PPD for the TP shall be determined by performing a series of at least three pours of the TP using its assessed charge weight, and calculating the average of the pours.

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4.5.5.3 Individual PPD determination. The PPD for an individual test pour shall be determined by measuring the maximum packing depth (MPD) and subtracting 0.7 inch from the MPD. Inert primer hardware shall be installed in the case to accurately represent displacement. The following shall be performed to determine the MPD for an individual test pour:

a. Pour the propellant charge steadily from a height of 30 inches above the mouth over an interval of 60 seconds, while the outside of the case is briskly tapped with a soft mallet adjacent to the rising level of the propellant during pouring.

b. Level the resulting propellant bed within the cartridge case.

c. Measure the distance (inches) from the mouth of the cartridge case to the top of the propellant bed.

4.5.6 Ballistic testing. Four types of ballistic tests shall be conducted for first article inspection and conformance inspection:

a. Charge weight probe tests (see 4.5.6.2.1).

b. Powder proof firings test (see 4.5.6.2.2).

c. Charge weight validity test (see 4.5.6.2.3).

d. Propellant interface test (see 4.5.6.3).

Propellant interface testing may only be omitted from the conformance inspection if approved by NAVSEA, as specified (see 6.2).

4.5.6.1 Test facility, guns, instrumentation, and hardware. Unless otherwise specified (see 6.2), the test facility, guns, instrumentation, and hardware shall be as described in the following subsections.

4.5.6.1.1 Test facility. The test facility shall be acceptable to the Government. Point of contact information at the test facility shall be as specified (see 6.2).

4.5.6.1.2 Gun and barrel.

4.5.6.1.2.1 Gun barrel. All ballistic tests shall be conducted using Fleet-released, 5"/54 or 5"/62 gun barrels (such as the MK 19, MK 36, or similar, as specified [see 6.2]) that have met the acceptance criteria during barrel proofing and currently do not exhibit a velocity loss greater than 50 ft/s in accordance with Drawing 2814087. The chamber volume resulting from the projectile and propelling charge combination shall be representative of that used in the Fleet. Instrumented gun barrel and cartridge case requirements can be found in 4.5.6.1.3.3 and 4.5.6.1.3.4, if required.

4.5.6.1.2.2 Gun mount. The test facility shall use an automatic gun mount (such as the MK 45, or similar, as specified [see 6.2]) and an open gun mount (such as the MK 39, or similar, as specified [see 6.2]) equipped with a barrel as described in 4.5.6.1.2.1 for all ballistic tests.

4.5.6.1.3 Instrumentation. The test facility shall use the following instrumentation. Additional instrumentation may be used at the discretion of the Government for information-only purposes, but shall not jeopardize the validity of the test data obtained with the instrumentation listed below.

4.5.6.1.3.1 Pressure instrumentation.

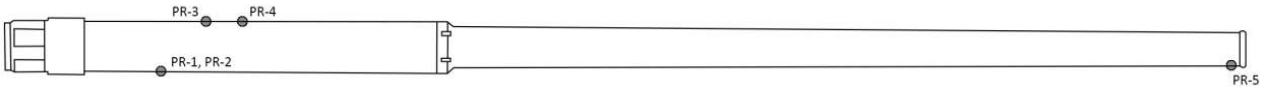
4.5.6.1.3.1.1 Copper crusher pressure gauges. Unless otherwise specified (see 6.2), the test facility shall determine the maximum pressure by use of three M11 copper crusher gauges installed in the base of the propelling charge. The gauges may be installed in holders secured to the base of the primer tube or placed in the bottom of the case in a vertical position. The copper spheres used in the M11 gauges shall be from a population that has been characterized by the Aberdeen Proving Ground (APG), or other facility approved by the Government, as specified (see 6.2). Tarage Tables specific to the lot of copper spheres shall be used to determine the pressure based on the amount of compression of the copper. The reported value shall be the average of the three gauge readings.

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4.5.6.1.3.1.2 Piezoelectric pressure gauges. If piezoelectric pressure gauges are used, the gauges shall be specifically designed for applications involving high pressures and very high instantaneous combustion temperatures. The gauges shall be calibrated prior to each test series via linear least squares regression of the pressure versus voltage. A charge amplifier system may also be necessary in order to convert the charge signal produced by the piezoelectric gauge into a low impedance voltage signal.

4.5.6.1.3.2 Velocity instrumentation. The test facility shall determine the muzzle velocity using an accurate measurement system. If a radar system operating on the Doppler principle is used, it shall be mounted and located behind the gun and be capable of transmitting within the X band frequency between 8 and 12 gigahertz (GHz). A Doppler radar analyzer shall be utilized for data acquisition and post processing that is capable of storing the results in its non-volatile memory.

4.5.6.1.3.3 Instrumented barrel. When an instrumented barrel is required, pressure gauges shall be located in accordance with figure 1. Prior approval from the technical DA shall be obtained for deviations from the locations on figure 1, or when a different barrel is desired, as specified (see 6.2).



Pressure Port Number	5"/54 BARREL		5"/62 BARREL	
	Distance from Breech Face (inches)	Position in Degrees [1/]	Distance from Breech Face (inches)	Position in Degrees [1/]
PR-1	35.00	330.0	35.00	330.0
PR-2	35.00	150.0	35.00	180.0
PR-3	45.00	2.0	45.00	350.0
PR-4	52.75	0.5	55.00	2.0
PR-5	268.00	268.5	308.00	170.0

NOTE:
[1/] 0-degree location is at 12 o'clock position, aft-looking forward.

FIGURE 1. Instrumented barrel pressure port locations.

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4.5.6.1.3.4 Instrumented cartridge case. When an instrumented cartridge case is required, pressure gauges shall be located in accordance with figure 2. Prior approval from the technical DA shall be obtained for deviations from the locations on figure 2, as specified (see 6.2).

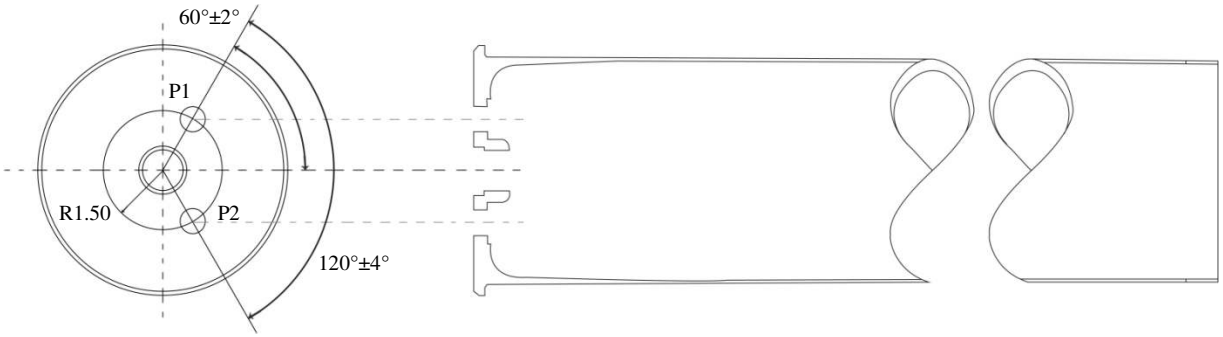


FIGURE 2. Instrumented cartridge case pressure port locations.

4.5.6.1.3.5 Eject time instrumentation. The test facility shall determine the eject time for each shot during all ballistic firing tests by recording the times at which the firing signal and muzzle exit occur and calculating the difference between them.

4.5.6.1.4 Test hardware.

4.5.6.1.4.1 Warming or conditioning rounds. The test facility shall fire warming or conditioning rounds to stabilize gun temperature and eliminate the biasing effects of firing the propellant at various charge levels or temperatures. The first round fired in a gun during a test is the warming round. The round fired as part of a temperature conditioning group or charge level group wherein the velocity is expected to differ from the previous group by approximately 100 ft/s or more is a conditioning round. Warming and conditioning rounds may have unusual ballistics, and the results, therefore, should be disregarded.

4.5.6.1.4.2 Propelling charges.

4.5.6.1.4.2.1 Assembly. The test facility shall assemble propelling charges to comply with the requirements of Drawing 3208375. The propelling charges shall be crimped, though the use of adhesive on the plug and cartridge case interface is not required for the charge weight probe (see 4.5.6.2.1) or powder proof firing (see 4.5.6.2.2) tests. Adhesive shall be used for the charge weight validity (see 4.5.6.2.3) and propellant interface (see 4.5.6.3) tests. Propelling charges shall be loaded to yield the packing depth within 0.4 inch of that determined in 4.5.5.2 for the charge weight to be fired.

4.5.6.1.4.2.2 Hardware. The hardware shall consist of sub-components (including cartridge cases, primers, discs, closure plugs, and wads) that shall be from one acceptable lot per component.

4.5.6.1.4.2.3 Conditioning. The cases, primers, and propellant for all propelling charges shall be conditioned at 70±3 °F for a minimum of 4 days immediately prior to assembly. The propelling charges used in the:

a. Charge weight probe test (see 4.5.6.2.1) shall be fired at 70±3 °F.

b. Powder proof firing test (see 4.5.6.2.2) shall be fired at 70±3 °F.

c. Charge weight validity test (see 4.5.6.2.3) shall be conditioned to 70±3 °F for at least 48 hours after assembly to allow the adhesive to cure thoroughly. They shall be fired at 70±3 °F.

d. Propellant interface test (see 4.5.6.3) shall be conditioned to 70±3 °F for at least 48 hours after assembly to allow the adhesive to cure thoroughly. Then they shall be conditioned for a minimum of 72 hours at temperatures of 20 °F, 50 °F, and 120 °F, and fired within 5 minutes of being removed from their conditioning chambers.

4.5.6.1.4.3 Projectiles. Projectiles shall be 70.0±0.1 pounds and inert loaded. The projectile bodies shall come from a single production lot.

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4.5.6.1.4.4 RP. The RP shall be as specified (see 6.2).

4.5.6.2 Charge assessment. The charge assessment shall consist of the following series of tests and calculations:

a. Charge weight probe test (see 4.5.6.2.1).

b. Powder proof firing test (see 4.5.6.2.2).

c. Matching charge evaluation calculations (see 4.5.6.2.2.3).

d. Velocity uniformity calculations (see 4.5.6.2.2.5).

e. Assessed charge calculations (see 4.5.6.2.2.6).

f. Charge weight validity test (see 4.5.6.2.3).

4.5.6.2.1 Charge weight probe test. Using instrumented cases and instrumented barrel (see 4.5.6.1.2 and 4.5.6.1.3) in the open gun mount and warming and conditioning rounds as needed (see 4.5.6.1.4.1), the following series of tests shall be conducted to:

a. Determine the mean velocity for the RP (see 4.5.6.2.1.2).

b. Determine the estimated TP charge weight needed to obtain a mean velocity that matches the obtained RP velocity that will be used during the powder proof firings (see 4.5.6.2.1.3).

c. Verify the estimated chamber pressure of the estimated TP charge weight is below the specified limit of 55,100 (+1,350) psi.

The upper tolerance of +1,350 psi shall only apply to the charge weight probe test.

4.5.6.2.1.1 Pressure, velocity, and eject time data. Pressure, velocity, and eject time data shall be determined for the charge weight probe tests (see 4.5.6.1.3).

4.5.6.2.1.2 Reference velocity. The test facility shall determine the mean velocity of the RP by firing at least three RP rounds. The RP rounds shall be loaded in accordance with 4.5.6.1.4.2 using the previously established charge weight for the RP, as specified (see 6.2). The average velocity of the RP rounds shall then be calculated.

4.5.6.2.1.3 Estimated charge weight. The test facility shall determine an estimated charge weight for the TP by firing TP rounds where the charge weight and PPD are adjusted to vary projectile muzzle velocity and chamber pressure. A sufficient number of TP rounds shall be fired to provide at least two rounds with velocities below the mean velocity for the RP, and at least two rounds with velocities at or above the mean RP velocity. The charge weight for the TP shall be calculated using a linear least squares regression of the charge weight versus velocity.

4.5.6.2.1.4 Rejection criteria. An estimate shall be made from the TP velocity versus charge weight data for how much propellant is required to match the mean RP projectile muzzle velocity (see 3.8.1). If the estimated maximum chamber pressure for the estimated TP charge weight exceeds 55,100 (+1,350) psi, the probing program shall be repeated. If the repeated probing program also has an estimated maximum chamber pressure that exceeds 55,100 (+1,350) psi, the lot of propellant shall be rejected without further testing.

4.5.6.2.2 Powder proof firings test. A test series shall be conducted in each of four gun barrels (see 4.5.6.1.2.1) in the automatic mount (see 4.5.6.1.2.2) to evaluate the performance of the TP against the performance of the RP. The final charge weight for the TP to be production loaded, referred to as the AC, shall be calculated (see 4.5.6.2.2.6).

4.5.6.2.2.1 Velocity and eject time data. Velocity and eject time data shall be determined for the powder proof firings, and pressure data shall be provided for informational purposes (see 4.5.6.1.3).

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4.5.6.2.2.2 Charges. Warming rounds shall be fired to stabilize the gun temperature (see 4.5.6.1.4.1). Three charges at each of five charge levels shall be fired in each gun barrel:

a. The RP charge shall contain the previously established charge weight for the RP, as specified (see 6.2).

b. The first TP charge (TC1) shall be the estimated TP charge weight determined in the charge weight probe test of 4.5.6.2.1.b.

c. The second TP charge (TC2) shall contain approximately 0.5 pound less TP than TC1.

d. The third TP charge (TC3) shall contain approximately 0.5 pound more TP than TC1.

e. The fourth charge, or Matching Charge (MC), shall be the charge weight of the TP predicted to match the average muzzle velocity of the RP from that gun on that day (see 4.5.6.2.2.3).

4.5.6.2.2.3 MC evaluation. A linear least squares equation shall be calculated for the velocity versus charge weight data for each gun. The MC shall be determined by substituting the required velocity in the linear least squares equation and solving for the charge weight. Three MC rounds shall be fired in each of the four guns to determine the MC projectile muzzle velocity.

4.5.6.2.2.4 Firing sequence. At a minimum, the rounds shall be fired in the following sequence in each of the four different gun barrels to obtain the data required from the powder proof firings test (see 4.5.6.2.2.2):

a. Two warmer rounds

b. TC2, TC2, TC2, RP, TC1, TC1, RP, TC1, RP, TC3, TC3, TC3

c. MC, MC, MC

4.5.6.2.2.5 Velocity variation. The velocity standard deviation (VSD) shall be determined from the MC data obtained from each gun barrel collected during the powder proof firings test (see 4.5.6.2.2.3).

4.5.6.2.2.5.1 VSD calculation. The following steps shall be used to determine the VSD:

a. Calculate the sum of the squares of the deviations from the mean for each gun.

b. Sum across the guns.

c. Divide by the quantity of the sum of the total number of rounds minus the number of guns.

d. Take the square root to the nearest unit of acceptance criteria.

4.5.6.2.2.5.2 VSD formula. The following formula (derived from the written steps in 4.5.6.2.2.5.1) shall be used to determine the VSD:

$$VSD = \sqrt{\frac{\sum_{i=1}^{4}\left(\sum_{j=1}^{n}\left(v_j - \bar{v}_i\right)^2\right)}{ROUNDS_{total} - GUNS_{total}}}$$

where:

$\bar{v}_i$ is the average velocity of the MC for the ith gun

v_j is the velocity of the jth MC for the ith gun

n is the number of MCs for the ith gun

4.5.6.2.2.5.3 Rejection criteria. The lot shall be rejected if the VSD does not meet the requirements of 3.8.3.

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4.5.6.2.2.6 Assessed charge. The following formula shall be used to determine the AC:

$$AC = \frac{\sum_{i=1}^{4}\left(\frac{MC_i}{\Delta_i^{\,2}}\right)}{\sum_{i=1}^{4}\left(\frac{1}{\Delta_i}\right)^2}$$

where:

MC_i is the charge weight of the TP estimated to match the mean RP projectile velocity in the ith gun

Δ_i is the difference between the mean muzzle velocity obtained for the matching charge in the ith gun (MC_i) and the mean RP projectile muzzle velocity in the ith gun

Note: If Δ_i is zero, replace the zero with 1.

4.5.6.2.3 Charge weight validity test. Using instrumented cases and instrumented barrel (see 4.5.6.1.2 and 4.5.6.1.3) in the open gun mount and warming rounds as needed (see 4.5.6.1.4.1), the test facility shall conduct a series of tests to determine the following:

a. Validity of the AC calculated in 4.5.6.2.2.6, and

b. Pressure variation calculated in 4.5.6.2.3.3.2.

4.5.6.2.3.1 Pressure and eject time data. M11 copper crusher gauges shall be used to obtain pressure data for the charge weight validity test. Eject time shall also be determined. Velocity data and pressures obtained from piezoelectric gauges shall be provided for informational purposes (see 4.5.6.1.3).

4.5.6.2.3.2 AC validity. The validity of the AC shall be determined by firing 10 RP rounds and 10 AC TP rounds in any one of the gun barrels used in the powder proof firings (see 4.5.6.2.2). The rounds shall be assembled as described in 4.5.6.1.4.2 and shall be randomized to eliminate any gun erosion or hot gun effects. The number of RP rounds can be negotiated based on DA consent, as specified (see 6.2).

4.5.6.2.3.3 Pressure variation. The maximum chamber PSD shall be determined from the same set of rounds used to determine the validity of the AC (see 4.5.6.2.3.2).

4.5.6.2.3.3.1 PSD calculation. The following steps shall be used to determine the PSD:

a. Calculate the sum of the squares of the deviations from the mean of the TP rounds.

b. Divide by the quantity of the sum of the total number of rounds minus 1.

c. Take the square root to the nearest unit of acceptance criteria.

4.5.6.2.3.3.2 PSD formula. The following formula (derived from the written steps in 4.5.6.2.3.3.1) shall be used to determine the PSD for the maximum pressures:

$$PSD = \sqrt{\frac{\sum\left(P_i - \overline{P}\right)^2}{(n-1)}}$$

where:

$\overline{P}$ is the average maximum chamber pressure of the TP

P_i is the maximum chamber pressure of the ith TP round

n is the number of rounds

4.5.6.2.3.3.3 Rejection criteria. The lot shall be rejected if the AC mean maximum chamber pressure and PSD do not meet the requirements of 3.8.2 and 3.8.4, respectively.

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4.5.6.3 Propellant interface test. Using instrumented cases and instrumented barrel (see 4.5.6.1.2 and 4.5.6.1.3) in the open gun mount and warming and conditioning rounds as needed (see 4.5.6.1.4.1), the test facility shall conduct a series of tests to verify there is no adverse primer-propellant relationship that could cause ballistic performance problems.

4.5.6.3.1 Rounds. The rounds shall be assembled as described in 4.5.6.1.4.2. They shall be conditioned to three different temperatures in accordance with 4.5.6.1.4.2.3.

4.5.6.3.2 Pressure, velocity, and eject time data. Pressure, velocity, and eject time data shall be recorded for the interface test (see 4.5.6.1.3). The temperature coefficient and $-\Delta P_i$ for the TP shall be determined by the appropriate Government gun propelling charge DA, as specified (see 6.2).

4.5.6.3.2.1 Pressure waves. Differential pressures shall be obtained from simultaneous pressure versus time measurements taken at the breech face and just behind the base of the seated projectile. A negative differential pressure shall be defined as the condition when the chamber pressure of a weapon at the forward end of the chamber is higher than the pressure at the breech end at some time during the pressure versus time history. Differential pressures shall be evaluated by at least one of these methods:

a. Analyzing the firings for evidence of erratic performance, such as long ignition delays (hangfires in excess of 300 milliseconds), high negative differential pressures, inflections in the pressure-time trace occurring prior to peak pressure, high peak chamber pressures, and high muzzle velocity standard deviations.

b. Determining the maximum initial negative differential pressure ($-\Delta P_i$) during the initial occurrence and the maximum $-\Delta P_i$ during any later occurrence that shows a larger value than during the initial phase.

c. Calculating the reverse pressure gradients ($\Delta P/\Delta T$) from the pressure difference profiles.

Methods (a) and (b) describe the amplitude or magnitude of the pressure waves and are explained in the International Test Operations Procedure (ITOP) 4-2-504. Method (c) describes the frequency of the pressure waves and correlates better with the pressurization rate.

4.5.6.3.3 Firing sequence. A minimum of three RP rounds loaded to the assessed charge weight of the RP and nine TP rounds loaded to the AC from 4.5.6.2.2.6 are required. At least one warming round and three conditioning rounds (see 4.5.6.1.4.2.3) will also be needed. The following interface test sequence shall be performed:

a. Fire one warming round to stabilize the gun temperature.

b. Fire 20 °F rounds: one conditioning round followed by three TP rounds and conclude with one RP round.

c. Fire 50 °F rounds: one conditioning round followed by three TP rounds and conclude with one RP round.

d. Fire 120 °F rounds: one conditioning round followed by three TP rounds and conclude with one RP round.

4.5.7 CB testing. CB testing shall be performed to determine the propellant burning properties for use in IB calculations and to identify attributes of process, ingredients, and grain configuration for comparative purposes. CB testing shall be conducted on the propellant in accordance with MIL-STD-286, Method 801.1.2 or 804.1, except the pressure range achieved shall be between 70,000 psi and 100,000 psi, or the test shall be discarded. Prior written approval from the Government shall be obtained when CB testing using pressures outside of the 70,000 psi to 100,000 psi range is desired, as specified (see 6.2). Unless otherwise specified (see 6.2), the CB RP shall be the same lot of RP as is used for 4.5.6. The test results shall be used to calculate RF (see 4.5.7.5.1), RQ (see 4.5.7.5.2), RV (see 4.5.7.5.3), and Burn Rate (BR) (see 4.5.7.5.4).

4.5.7.1 Apparatus. The CB shall have a constant volume between 12 cubic inches (in^3) and 43 in^3 (200 cubic centimeters [cm^3] and 700 cm^3). The CB shall be equipped with firing electrodes to effect ignition, a pressure transducer, gas release valve, and thermocouple. It shall be surrounded by a temperature controlled water jacket to maintain the required test temperature.

4.5.7.2 Pressure gauge. Measurement of the pressure shall be achieved using a piezoelectric pressure transducer mounted in an adapter or other stress free environment. The minimum frequency response of the pressure transducer shall be 25 kilohertz (kHz). A valid calibration shall be required prior to firing using a linear or polynomial fit of the observed electrical output and pressure input accurate to within 0.1 percent.

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4.5.7.3 Ignition system. The ignition system shall consist of a power source and the igniter material. The power source shall greatly exceed the minimum energy requirements for consistent ignition of the electric match, squib, or the igniter material. The igniter material shall be black powder, clean burning igniter, or Benite strands. An alternate ignition system may be used upon approval by the Government, as specified (see 6.2).

4.5.7.4 Operation. The propellants shall be conditioned at the test temperature ±2 °F for at least 8 hours. The humidity level and temperature of the RP shall be controlled and monitored. A warming shot shall be fired prior to the start of testing, followed by at least three firings of both the RP and TP samples. The RP and TP shots should alternate, and the CB shall be swabbed after each shot. The test series shall be completed in one shift. If any test series is interrupted for more than 2 hours, another warming shot may be required.

4.5.7.5 Data. The data reported from the CB testing shall include the experimental maximum pressure and pressure versus time curve (dP/dt). The dP/dt shall be obtained analytically from the pressure versus time data. The CB test results shall be used to calculate RF (see 4.5.7.5.1), RQ (see 4.5.7.5.2), RV (see 4.5.7.5.3), and BR (see 4.5.7.5.4).

4.5.7.5.1 RF. RF shall be defined as the ratio of the maximum pressures developed between the TP and the RP when tested at the same loading density and initial temperature in the same CB. The formula for RF is provided below for convenience; verification of its currency and example calculations are included in MIL-STD-286, Method 801.1.2 and 804.1.

$$RF = \frac{100}{m}\sum_{i=1}^{m}\left(\frac{T_{p\max,i}}{R_{p\max,i}}\right)$$

where:

m is the number of reference or test firings

$T_{p\ max,\ i}$ is the test propellant maximum pressure for i^{th} firing

$R_{p\ max,\ i}$ is the reference propellant maximum pressure for i^{th} firing

4.5.7.5.2 RQ. RQ shall be defined as the ratio of dP/dt of the TP to the dP/dt of the RP taken at specific pressure intervals at the same initial temperature and loading density in the same CB. The formula for RF is provided below for convenience; verification of its currency and example calculations are included in MIL-STD-286, Method 801.1.2 and 804.1.

$$RQ = \frac{1}{m}\sum_{i=1}^{m}\left(\frac{100}{n}\sum_{j=1}^{n}\frac{T_{ij}}{R_{ij}}\right)$$

where:

m is the number of reference or test firings

n is the number of data points per firing

T_{ij} is the dP/dt test propellant value of ith firing and j^{th} data point

R_{ij} is the dP/dt reference propellant value of ith firing and j^{th} data point

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4.5.7.5.3 RV. RV is defined as the average ratio of TP vivacity to RP vivacity determined from nine equally spaced increments within the normalized pressure range of 0.20 (P/P_{max}) to 0.60 (P/P_{max}). Of the nine TP to RP vivacity ratios, the value furthest from unity (value of 1) shall be ignored. The RV shall be calculated from the eight remaining points.

4.5.7.5.3.1 Vivacity. Vivacity is an expression of the propellant energy release that is proportional to the mass rate of combustion. The mass rate of combustion, and subsequent gas generation, at any point in the combustion process is dictated by two factors:

a. The instantaneous, pressure dependent, burning rate of the propellant primarily driven by the chemistry of the propellant, and

b. The amount of propellant available for combustion primarily driven by the physical geometry of the propellant grain.

4.5.7.5.3.2 Vivacity calculation. The equation for determining the vivacity is as follows:

$$Vivacity = \frac{dP(t)/dt}{P(t) * P_{max}}$$

where:

dP(t)/dt is the time derivative of pressure

P(t) is the pressure

P_{max} is the maximum pressure

Plots of vivacity versus normalized pressure (P/P_{max}) shall be provided for future use in IB calculations to predict initial RP charge weight and TP performance.

4.5.7.5.4 BR. BR is a calculated value based on the grain geometry, dimensions, density, and thermochemical properties of the propellant, such as impetus, co-volume, flame temperature, and the ratio of specific heats. When the propellant burns in a manner consistent with its granulation as shown in the vivacity curve, the BR can be determined. A value for the BR shall be calculated using a BR reduction program, such as NSWC IHEODTD's CBBR or APG's BRLCB or XLCB, and reported for future use in IB modeling.

4.5.8 IB modeling. IB modeling shall be performed to predict the estimated charge weight required for use in the charge assessment described in 4.5.6.2. At a minimum, gun and barrel specifics from 4.5.6.1.2, primer and projectile details from 4.5.6.1.4.2 and 4.5.6.1.4.3, propellant characteristics from 4.5.1 and 4.5.2, and BR from 4.5.7.5.4 shall be used as inputs into the model. IB modeling shall be performed using a modeling code, such as NGEN, IBHVG2, IBCODE, FNGUN, XKTC, INTBAL, or other similar code, as specified (see 6.2), and deemed validated by the DA.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.1.1 Ballistic testing and performance evaluation. The propellant covered by this specification is intended for use in the MK 65 clearing charge and MK 67 propelling charge, which are used in MK 45 gun systems. The ballistic performance requirements contained within this specification pertain to the MK 67 propelling charge only and should not be applied to the MK 65 clearing charge or MK 68 reduced charge without prior review of historical performance data or IB modeling predictions for charge weight and muzzle velocity at a minimum.

First article and conformance inspection testing under this specification can be completed with either the 5"/54 weapon system (utilizing the MK 45 Mod 1 or 2) or the 5"/62 weapon system (utilizing the MK 45 Mod 4), with the only difference in requirements being the velocity variation.

6.1.2 Propellant. NACO is the designation for Navy “cool” low-force, low-flame temperature, single-base propellant types that contain low-nitration nitrocellulose (12.0 percent nitrogen, nominal). The type covered by this specification also contains n-Butyl Stearate as a coolant and potassium sulfate as a flash suppressant, and is designated BS-NACO. The propellant was designed for military use and has no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. If required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2).

c. If first article is required and, if so, relevant information including sample size, sample origin, examination/methods, location, approval, and disposition (see 3.1, 4.2, 4.3.1, and 6.3.1).

d. Requirement to determine average nitrogen content if requested (see table I).

e. Relevant information for conformance inspection, including sample size, sample origin, examination/methods, location, approval, and disposition (see 4.3).

f. Address and contact information for the U.S. Navy Gun Ammunition Propelling Charge Design Agent (DA) and the U.S. Navy Major Caliber Propelling Charge Acquisition Engineering Agent (AEA) (see 4.3, 4.5.6.3.2, 6.4.1, and 6.4.2).

g. Sample submission and lot size, if other than specified (see 4.3.1).

h. Activity to conduct gun performance test (see 4.3.2.1, 4.5.6.1.1, and 6.4.1).

i. Address and contact information for physical characteristics testing facility (see 4.3.2.2).

j. Address and contact information for U.S. Navy’s surveillance activity for propellants (see 4.3.2.3).

k. Certification requirements related to chemical composition (see 4.5.1).

l. Alternative test methods for determining chemical composition, when appropriate (see 4.5.1).

m. Circumstances or criteria required for early termination of ballistic testing or lot rejection based on preliminary PPD during first article inspection, if necessary (see 4.5.5).

n. When propellant interface testing may be omitted from conformance inspection (see 4.5.6).

o. Guns, instrumentation, and hardware to be used for ballistic testing, if other than specified (see 4.5.6.1).

p. Gun barrel to be used if different than MK 19 or MK 36 (see 4.5.6.1.2.1).

q. Automatic gun mount to be used if different than MK 45 and open gun mount to be used if different than MK 39 (see 4.5.6.1.2.2).

r. Pressure instrumentation, if other than specified (see 4.5.6.1.3.1.1).

s. Approved facility for pressure instrumentation characterization if other than APG (see 4.5.6.1.3.1.1).

t. Instrumented barrel pressure gauge locations and barrel used, if other than specified (see 4.5.6.1.3.3).

u. Instrumented cartridge case pressure gauge locations, if other than specified (see 4.5.6.1.3.4).

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v. RP used for testing (see 4.5.6.1.4.4).

w. Charge weight for the RP (see 4.5.6.2.1.2, 4.5.6.2.2.2, and 6.3.3).

x. Circumstances or criteria required for firing fewer RP rounds, if necessary (see 4.5.6.2.3.2).

y. Direction on whether to require propellant interface testing for conformance inspection (see 4.5.6.3.2).

z. CB testing using alternate RP or pressures outside of 70,000 psi to 100,000 psi range, if applicable (see 4.5.7).

aa. RP to be used for CB if different than the RP used for 4.5.6 (see 4.5.7).

bb. Variant ignition system for closed bomb testing, if applicable (see 4.5.7.3).

cc. Additional IB codes acceptable to the Government, if necessary (see 4.5.8).

dd. Packaging requirements and special markings (see 5.1).

ee. Request for alternate bids, if applicable (see 6.3.1).

ff. Requirements for gun propellant description sheets (see 6.4.1).

gg. Requirements for gun performance test reports (see 6.4.2).

hh. Address and contact information for the Government contracting officer (see 6.4.1 and 6.4.2).

ii. Sections of 49 CFR specifying the packing requirements (see 6.6.1).

jj. Sections of 49 CFR specifying the marking requirements (see 6.6.2).

6.3 Technical guidance.

6.3.1 Technical guidance for first article. First article inspection is intended to verify the propellant manufacturer has the manufacturing processes and quality control suitable for producing BS-NACO for military use. It can also be used to provide information needed to establish or validate the propellant granulation to be used during full-scale production. When a first article inspection is required, as specified (see 6.2), the contracting officer should provide offerors specific guidance as to whether the propellant to be used as the first article inspection sample should:

a. Be selected from the preliminary production of the first production lot, or

b. Come from dedicated first article pilot lots for establishing propellant granulation (see 4.2.1).

The quantity of propellant required for testing is specified in 4.2.1. The contracting officer should also include specific instructions in the acquisition documents regarding arrangements for:

a. Examination of first article samples,

b. Approval of first article test results, and

c. Disposition of first article samples, as specified (see 6.2).

The Government reserves the right to waive the requirement of first article inspection samples for bidders offering a product that has been previously acquired or tested by the Government. Bidders offering such products who desire to rely on such production or tests, must supplement their bid with proof that prior approval by the Government is applicable and appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation (see 6.2).

6.3.2 Technical guidance for projectile muzzle velocity. The charge weight of the TP should be assessed so the resulting nominal average projectile muzzle velocity matches that obtained for the RP via the powder proof firings (see 4.5.6.2.2 and 4.5.6.2.2.3). The following is provided to assist in choosing a starting point for how much propellant is required to match the performance of the RP for a MK 67 propelling charge in the 5”/54 gun system:

a. The nominal average projectile muzzle velocity has been 2,650 ft/s in a 5"/54 gun system.

b. Historical data shows conservative charge weights of 19.5 lbs.

6.3.3 Technical guidance for RP mean velocity. The charge weight of the RP has already been determined through prior charge assessment testing via the powder proof firings. It can be found in the lot acceptance test (LAT) report from that testing and may also be found on the relevant ammunition data card.

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6.4 Description sheets and test reports.

6.4.1 Propellant description sheets. The propellant manufacturer should furnish a propellant description sheet in accordance with MIL-STD-1171 giving a complete history of the manufacture of each propellant lot and the results of all tests performed on it. A copy of each description sheet should be submitted to each of the following:

a. Gun Propelling Charge DA
b. U.S. Navy Major Caliber Propelling Charge AEA
c. Activity designated to perform the gun performance tests
d. Government contracting officer

Addresses and point of contact information for these organizations should be included as specified (see 6.2).

6.4.2 Gun performance test reports. The facility that performs the gun performance tests (see 4.5.6.1) should furnish a complete test report, including all test, gun, projectile, and cartridge component data. A copy of each test report should be submitted to each of the following:

a. Gun Propelling Charge DA
b. U.S. Navy Major Caliber Propelling Charge AEA
c. Government contracting officer

Addresses and point of contact information for these organizations should be included as specified (see 6.2).

6.5 Summarized information.

6.5.1 Complete summary. Table IV contains a summary of most of the requirements called out in this specification in sections 3 and 4, along with the appropriate test methods and reference documents. It also reflects the suggested responsibilities in cases where the contractor does not have the capability to perform the required ballistic or closed bomb tests. Although it is for informational purposes only, the intent is to minimize the need for flipping back and forth within the various sections of this specification and to help outline the tasks that need to be performed for a better understanding of the BS-NACO propellant acceptance processes.

6.5.2 Summary of ballistic testing. Table V contains a summary of the ballistic tests required and suggests the firing sequences that should be used and the data that should be obtained for each type of test. The intent is to offer an idea of how much BS-NACO is required to complete the testing and other pertinent details necessary for test planning purposes.

6.5.3 Process flowchart. The process required for propellant acceptance of BS-NACO is depicted on figure 3. Although generalized, it shows how the information obtained from chemical and physical analysis and closed bomb testing is used to feed the IB models, which are then used to predict starting points for ballistic testing. The order of ballistic testing is also evident for planning purposes.

6.5.4 Authorization. The pilot lot test data will be reviewed by Government engineering and quality assurance personnel, upon which a granulation size will be selected that best fits the requirements of this specification. The contracting officer will be notified of the selected granulation and will communicate with the contractor to produce the selected granulation. If none of the pilot lots submitted meet the requirements of this specification, a new first article sample will need to be submitted in accordance with 4.3.1. All future production lots will be of the same basic granulation.

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TABLE IV. Summary of requirements and suggested responsibilities.

Characteristic	Spec Req	Spec Verification	Reference (or Test)	MIL-STD-286 (or Test Method)	Requirement	FAT	CI Testing
Composition	3.2	4.5.1			(Percent by Weight)	Government	Contractor
Nitrocellulose	3.2	4.5.1	MIL-DTL-244	209.2.1	93.55 (nominal)	Government	Contractor
n-Butyl Stearate	3.2	4.5.1	MIL-B-21465	222.1	3.00±0.30	Government	Contractor
Ethyl Centralite	3.2	4.5.1	MIL-E-255	202.2.3	1.20±0.20	Government	Contractor
Basic Lead Carbonate	3.2	4.5.1	MIL-DTL-18618	311.1.3 or 316.1	1.00±0.20	Government	Contractor
Potassium Sulfate	3.2	4.5.1	DTL-WS-35709	310.3.1, 310.4.2, or 316.1	1.25±0.30	Government	Contractor
Moisture	3.2	4.5.1	---	103.5	1.00 (min.) to 3.00 (max.)	Government	Contractor
Total Volatiles	3.2	4.5.1	MIL-STD-652	103.5	5.0 max.	Government	Contractor
Graphite	3.2, 3.4	4.5.1, 4.5.3	MIL-G-155	308.1.4	0.10±0.05	Government	Contractor
Grain Form and Dimensions	3.3	4.5.2, 4.5.3		504.1.1 or 504.6.1		Government	Contractor
Perforations	3.3.1	4.5.2, 4.5.3			7 perforations	Government	Contractor
Length	3.3.2	4.5.2, 4.5.3			2.10≤L/D≤2.50	Government	Contractor
Glaze	3.4	4.5.3	MIL-G-155	308.1.4	0.10±0.05	Government	Contractor
Integrity	3.5, 3.10	4.5.3	Visual			Government	Contractor
Stability	3.6	4.5.4		404.1.2	Methyl Violet Test Paper Self-Ignition	Government	Contractor
Production Packing Depth	3.7	4.5.5	Packing Depth Test		≥3.5 inches	Government	Government
Preliminary PPD	3.7	4.5.5.1	Packing Depth Test			Government	Government
Final PPD	3.7	4.5.5.2	Packing Depth Test			Government	Government

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TABLE IV. Summary of requirements and suggested responsibilities – Continued.

Characteristic	Spec Req	Spec Verification	Reference (or Test)	MIL-STD-286 (or Test Method)	Requirement	FAT	CI Testing
Gun Ballistic	3.8	4.5.6				Government	Government
Velocity	3.8.1	4.5.6.2.1	Charge Assessment	Charge Weight Probe	2,650 ft/s (nominal)	Government	Government
Pressure	3.8.2	4.5.6.2.1	Charge Assessment	Charge Weight Probe	≤55,100 psi	Government	Government
Velocity Variation	3.8.3	4.5.6.2.2.5	Charge Assessment	Powder Proof Firings	≤10 ft/s	Government	Government
Pressure Variation	3.8.4	4.5.6.2.3.3	Charge Assessment	Charge Weight Validity	≤1,700 psi	Government	Government
Temperature Coefficient	3.8.5.1	4.5.6.3	Propellant Interface		±0.2 (ft/s)/°F	Government	Government
Eject Time	3.8.5.2	4.5.6.3	Propellant Interface		≤50 ms	Government	Government
Pressure Waves	3.8.5.3	4.5.6.3	Propellant Interface		≤5,000 psi	Government	Government
Closed Bomb	3.9	4.5.7	Closed Bomb Testing			Government	Government
Relative Force	3.9.1	4.5.7.5.1	CB RF Calc		±2.0% of RP	Government	Government
Relative Quickness	3.9.2	4.5.7.5.2	CB RQ Calc		±3% of RP	Government	Government
Relative Vivacity 1/	3.9.3	4.5.7.5.3	CB Vivacity Plot		0.95-1.05	Government	Government
Burning Rate 1/	---	4.5.7.5.4	CB BR Software			Government	Government
Workmanship	3.10	4.5.3	Visual			Government	Government
IB Modeling 1/	---	4.5.8	Acceptable IB Code	NGEN, IBHVG2, etc.		Government	Government

NOTE:

1/ Inputs into the IB modeling code are needed to generate predictive starting charge weight values for ballistic testing.

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TABLE V. Ballistic test requirements per lot of test propellant.

Test Type	Spec Verification	Gun Mount	# of TP Rounds	# of RP Rounds	# of Warmer Rounds	# of Conditioning Rounds	Total # of Rounds	Firing Sequence	Data	When Req 1/
Ballistic Tests	4.5.6	At least 2,000 pounds of each lot of propellant is sufficient for ballistic testing of ~74 rounds TP								
Charge Weight Probe 2/	4.5.6.2.1	Open	~7	3	2	---	~12	2 warmers 3 rounds RP ~7 rounds TP	P (req) V (req)	FAT and CI
Powder Proof Firing 2/ 3/	4.5.6.2.2	Auto	12 (per barrel) 48 (total)	3 (per barrel) 12 (total)	2 (per barrel) 8 (total)	---	17 (per barrel) 68 (total)	2 warmers 3 rounds TC2 1 round RP 2 rounds TC1 1 round RP 1 round TC1 1 round RP 3 rounds TC3 3 rounds MC	P (info) V (req)	FAT and CI
Charge Weight Validity 4/ 5/	4.5.6.2.3	Open	10	10	2	---	22	2 warmers 10 rounds RP 10 rounds TP at AC	P (req) V (info)	FAT and CI
Propellant Interface 4/	4.5.6.3	Open	9	3	1	3	16	1 warmer 1 conditioning (20 °F) 3 rounds TP 1 round RP 1 conditioning (50 °F) 3 rounds TP 1 round RP 1 conditioning (120 °F) 3 rounds TP 1 round RP	P (req) V (req)	FAT Maybe CI 6/
Totals			**74**	**28**	**13**	**3**	**118**			

NOTES:

1/ FAT – First Article Inspection Testing or First Article Testing; CI – Conformance Inspection Testing.

2/ The use of adhesive on the plug/cartridge case interface is not required.

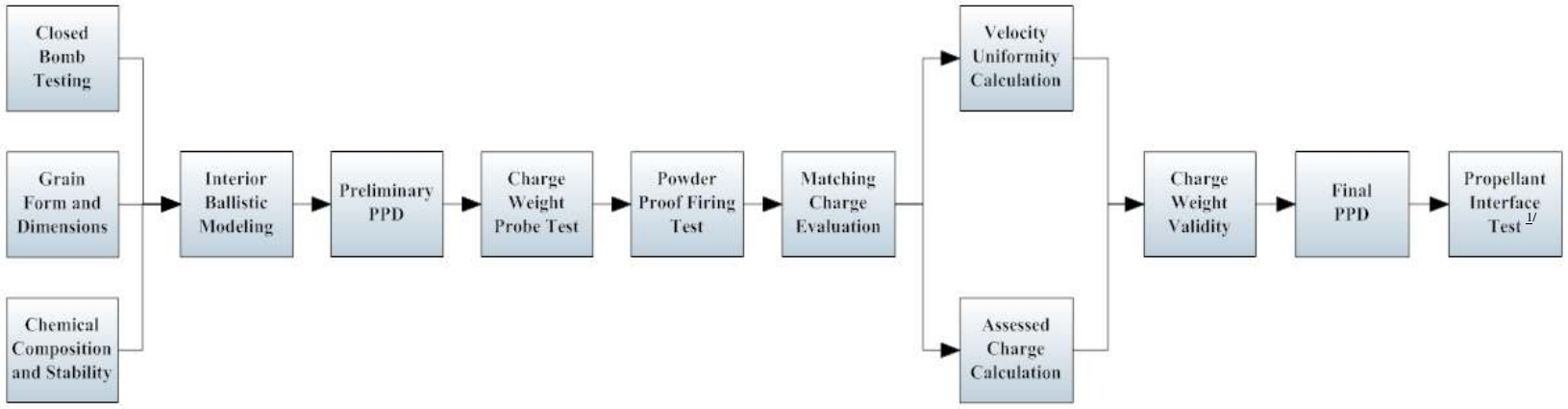
3/ Four separate tests are required using a different barrel for each test.

4/ The use of adhesive on the plug/cartridge case interface is required.

5/ Less RP rounds may be used if deemed sufficient by the DA.

6/ At the discretion of the Government.

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NOTE:

1/ At the discretion of the Government (see 4.5.6.3).

FIGURE 3. Process flowchart.

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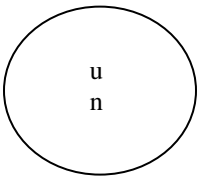
6.6 Packaging.

6.6.1 Packing. Packing should be at Level A. The propellant should be packed in MK 7 packing box containers in accordance with Drawing 138439 that are clean, dry, and free from foreign material. Immediately prior to packing, all containers should be tested with compressed air at an internal pressure of 2 to 3 psi. The containers should withstand this pressure for 15 seconds without leakage. The MK 7 packing box containers should also conform to the sections of 49 CFR, as specified (see 6.2). Palletizing should be in accordance with WR-53-724-1.

Fiberboard drum, in accordance with Army Drawing 12972488, is an acceptable alternate to the MK 7 container; palletizing in accordance with Drawing 7517077 should be used for this alternative. Note that the unit load should not be stacked in storage, transportation, or for any other purpose.

An Army M2 steel box, in accordance with Army Drawing 76-4-53, may be used to ship propellant samples under 20 pounds. Containers should be sealed using a metallic seal in accordance with Drawing 8794342 (Aluminum Seal).

6.6.2 Marking. In addition to any special markings required by the contract or order, shipping containers should be marked in accordance with the requirements of MIL-STD-129 and the sections of 49 CFR, as specified (see 6.2). The shipping name and serial number should be in accordance with 49 CFR, "POWDER, SMOKELESS, UN 0161". The United Nations (UN) Performance Oriented Packaging (POP) marking depicted below should be applied on the side of the container opposite the identification marking. The letters should be at least 0.50 inch high. The encirclement of the u/n symbol should be large enough to provide a minimum clear spacing around letters. The double asterisks should (**) indicate the year packed, e.g., 09 for 2009.



3A1/Y70/S/**

USA/DOD/NAD

6.6.3 Lot identification.

6.6.3.1 Propellant lot number. Propellant lot numbers should be in accordance with MIL-STD-1168. Note that bulk propellants are considered an exception to the lot numbering system for standard ammunition as defined in MIL-STD-1168 and should be constructed according to the unique provisions in MIL-STD-1168.

6.6.3.2 Propellant index. Upon successful completion of conformance inspection testing and acceptance of the propellant lot, the Navy propellant index should be assigned by the Conventional Ordnance Division, Code G2, at the NSWC IHEODTD, Picatinny Detachment.

6.7 Safety precautions. The safety precaution requirements of DoD 4145.26-M are applicable. The requirements of OP-5 are applicable if this specification is being used when a Government activity is performing the work. The safety precautions of AMCR-385-1 00 are applicable if an Army activity is performing work using this specification.

6.8 Acronyms. Additional acronyms and abbreviations used within this specification include:

5"/54	5-inch, 54-caliber
5"/62	5-inch, 62-caliber
AC	Assessed charge
AEA	Acquisition engineering agent
APG	Aberdeen proving ground
BR	Propellant burning rate
BS-NACO	Butyl Stearate Navy Cool
C	Critical characteristic

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CB	Closed bomb
CI	Conformance inspection testing
DA	Design agent
FAT	First article inspection or first article testing
IB	Interior ballistics
ITOP	International test operations procedure
LAT	Lot acceptance test
M	Major characteristic
MC	Matching charge
MPD	Maximum packing depth
NACO	Navy Cool
NSWC DD	Naval Surface Warfare Center, Dahlgren Division
NSWC IHEODTD	Naval Surface Warfare Center, Indian Head Explosive Ordnance Disposal Technology Division
P	Chamber pressure
POP	Performance oriented packaging
PPD	Production packing depth
PSD	Maximum chamber pressure standard deviation
RF	Relative force
RP	Reference propellant
RQ	Relative quickness
RV	Relative vivacity
TC1	First test propellant charge
TC2	Second test propellant charge
TC3	Third test propellant charge
TP	Test propellant
V	Muzzle velocity
VSD	Velocity standard deviation

6.9 Subject term (key word) listing.

5"/54 propelling charge

5"/62 propelling charge

Navy Cool

n-Butyl Stearate

MK 45 Mod 1 gun mount

MK 45 Mod 2 gun mount

MK 45 Mod 4 gun mount

MK 65 Mod 0, 5-inch, 54-caliber clearing charge

MK 67 Mod 3, 5-inch, 54-caliber propelling charge

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6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity:
Navy – OS
(Project 1376-2016-003)

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